

PROBLEM STATEMENT #62 | Sustainability & Green Tech

Food Bank Surplus Redistribution Platform

1. Problem Context & Overview

A food rescue platform where restaurants and grocery stores post surplus edible food batches, and verified local shelters reserve pickup windows before food expiry.

Target Stakeholders / Actors: *Donor Partner, Shelter Coordinator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.

Sample Acceptance Criteria: Pass: Shelter reserves batch and receives pickup verification code; Fail: Expired food batch remains active on claim board.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The platform shall dispatch instant push notifications to shelters within a 5 km radius when high-quantity perishable food is listed.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.